

Chronic, Wireless Recordings of Large Scale Brain Activity in Freely Moving Rhesus Monkeys

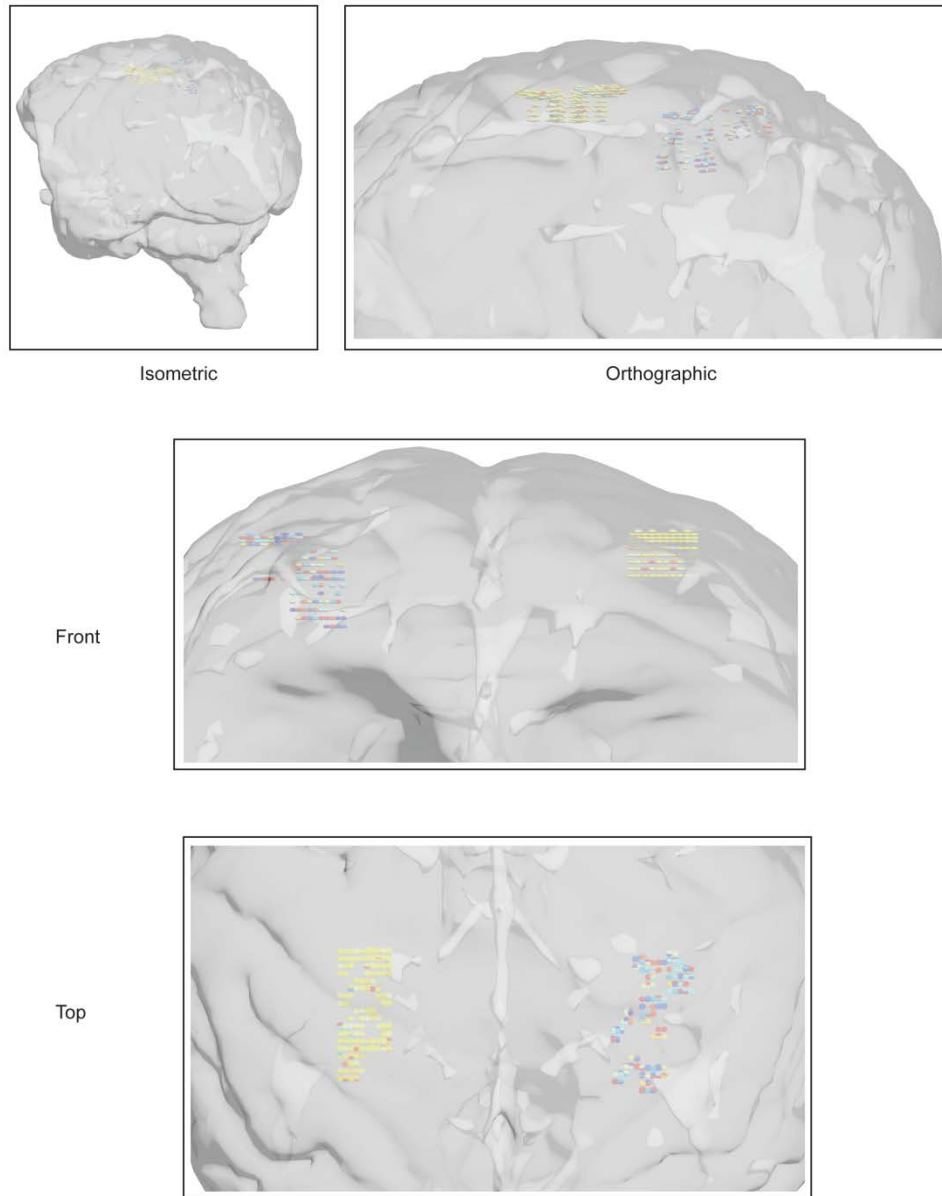
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Supplementary Material

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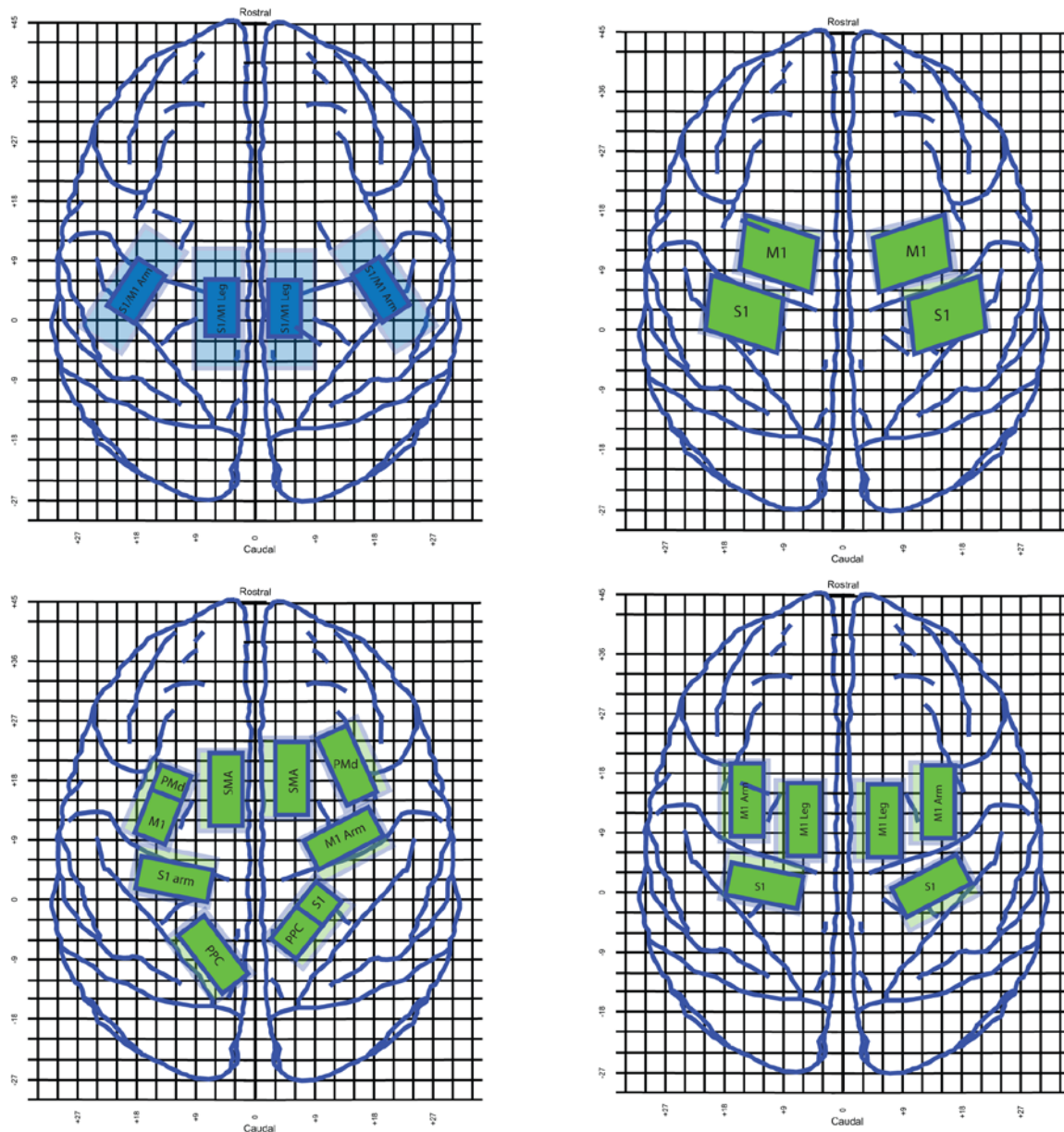
Supplementary Figures and Legends

Supplementary Figure 1 | Array density



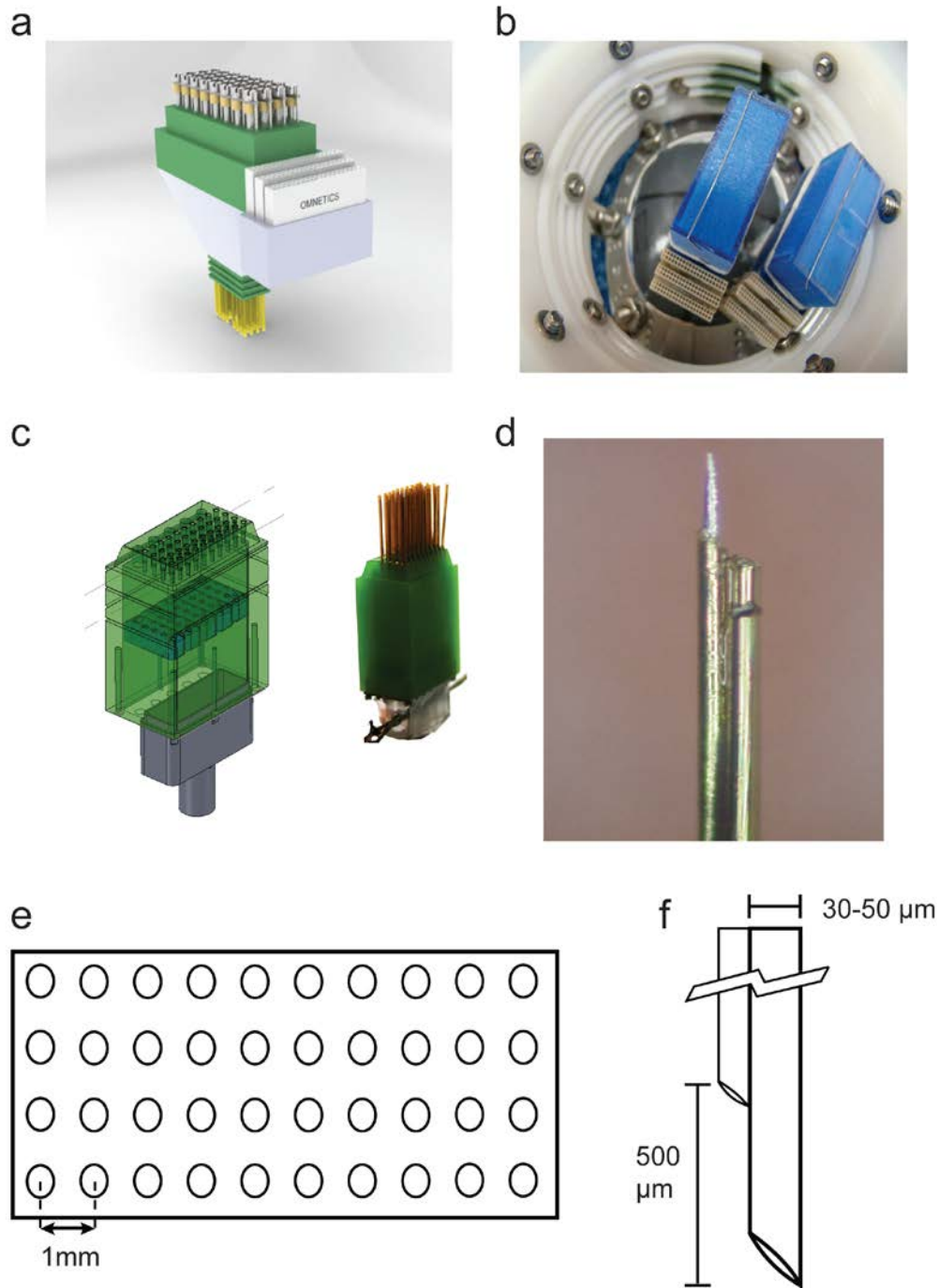
First panel is a three dimensional reconstruction of VLSBA array density at various angles for 560 recorded units. Each colored sphere is a single unit recorded over a simple reaching task. Positions are calculated by extrapolating the microwire contact from which the neuron was recorded and summing to the estimated position of the array in the somatomotor cortex. All panels show different views of the same models, as identified.

Supplementary Figure 2 | Implant locations



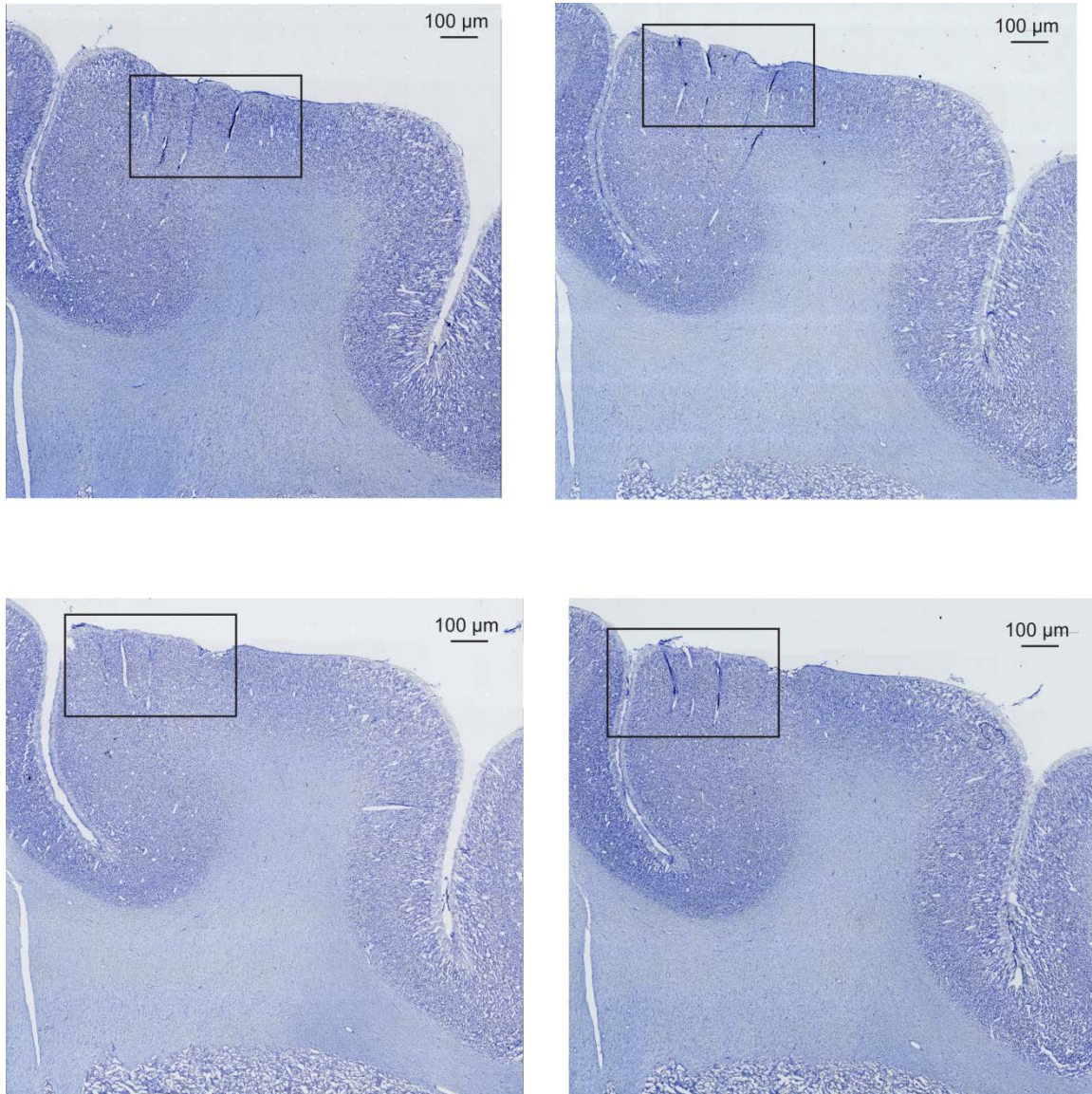
Implant locations for monkeys M, O, C and K implanted with first (top left), third (top right), and second generation (bottom row) recording cubes, respectively.

Supplementary Figure 3 | Previous generation arrays



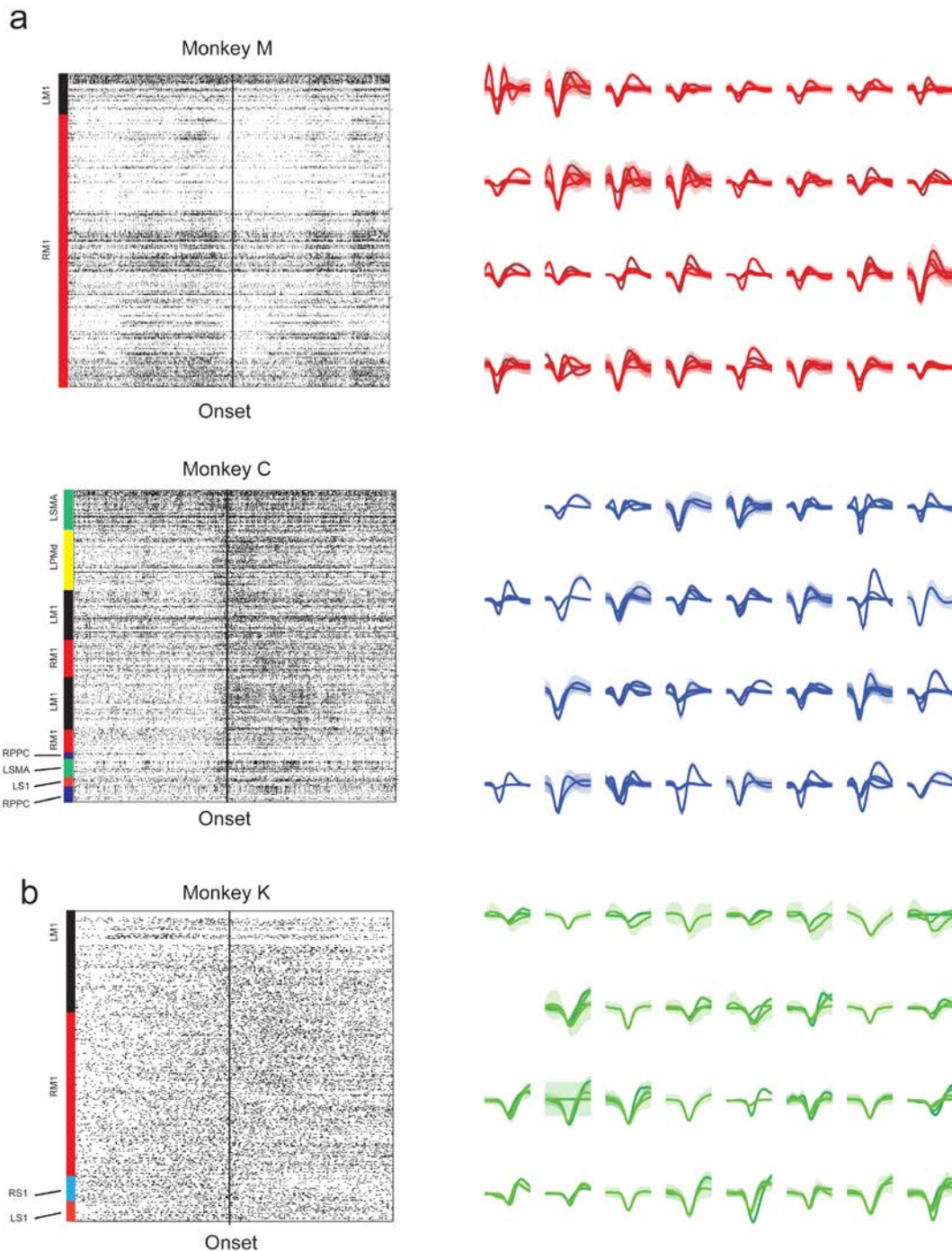
- (a) 1st generation array. Panel shows a 3D representation of a 96 cannula array.
 (b) A picture of these arrays mounted on a headcap previous to implantation.
 (c) 2nd generation is a 4x10 cannula array showing internal moving mechanism.
 (d) Photo of the bundled microwires (2nd gen) with guiding microelectrode.
 (e) Diagram of 4x10 array showing arrangement and spacing.
 (f) Diagram of bundle showing spacing between wires and diameter range.

Supplementary Figure 4 | Histology of electrode penetrations



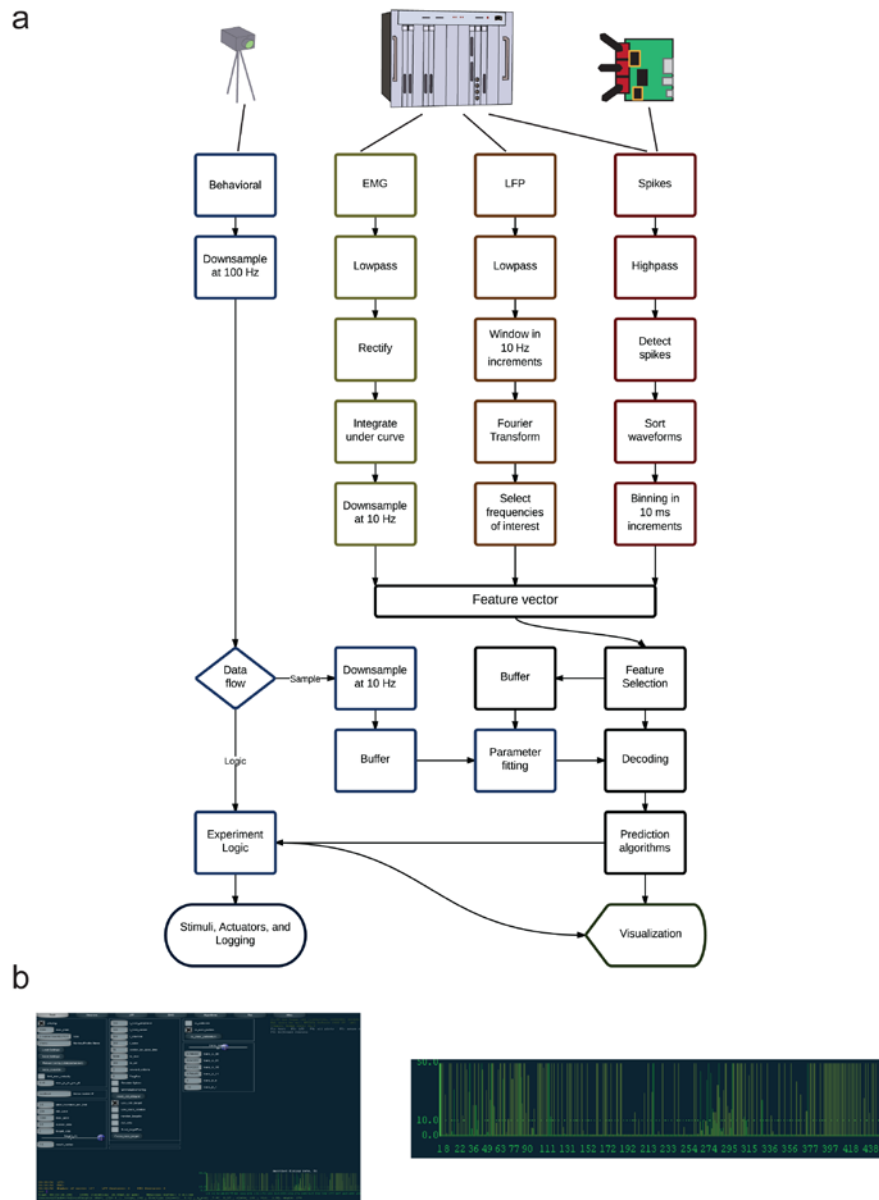
Histology shows tracts in the upper left quadrant of four slices of M1 cortex taken from Monkey T. Tracts are indicated by enclosing boxes.

Supplementary Figure 5 | Sample raster plots of recordings



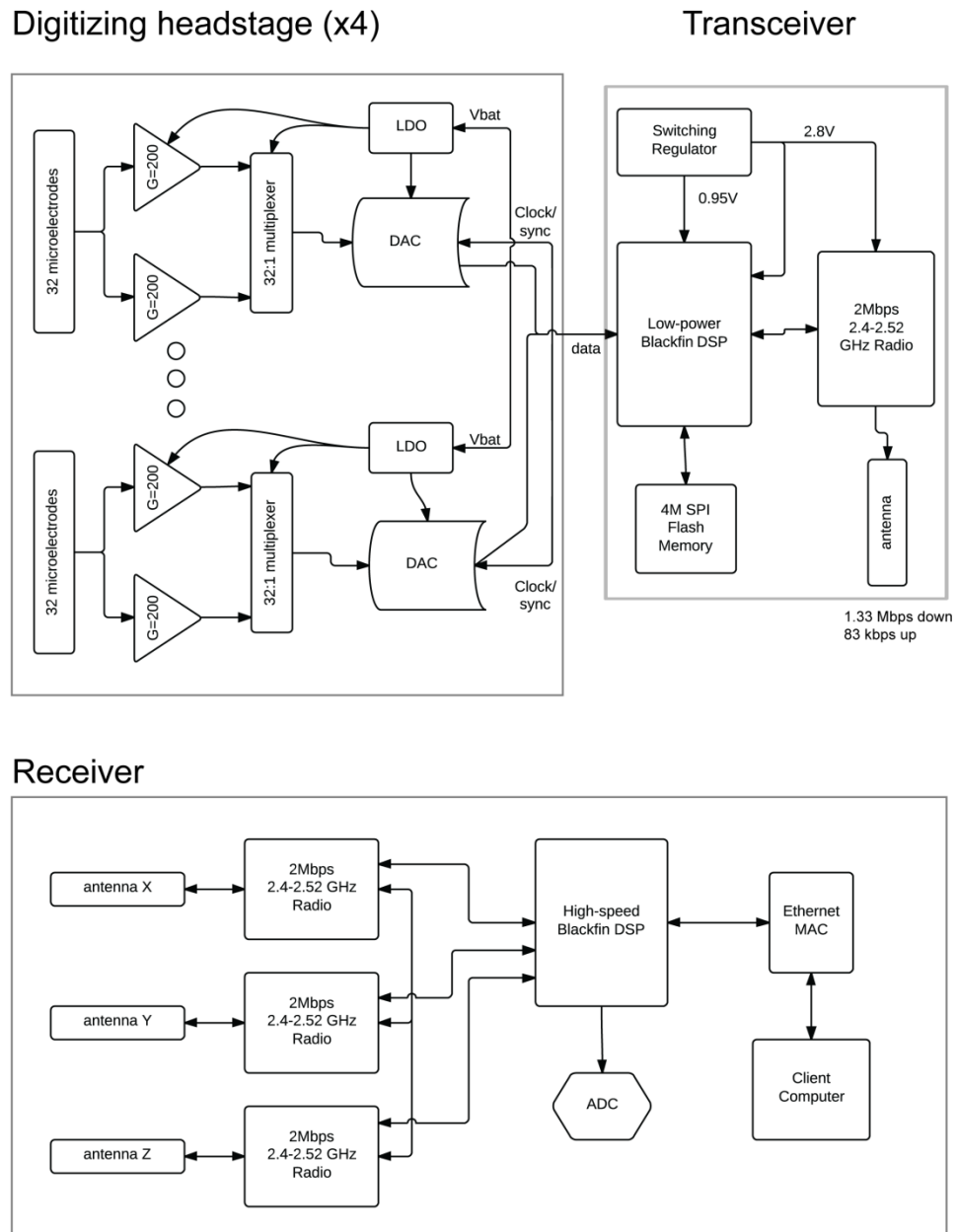
(a) Raster plots of monkeys C and M showing VLSBA activity during a 1 s time frame, centered on hand movement during a center-out task. Each row is an individual neuron identified by area with the color key in the y-axis. Onset of joystick movement is marked with a vertical bar. **(b)** Raster plot of monkey K performing a locomotor task over a 0.5 s time frame. Onset of swing phase is marked with a vertical bar.

Supplementary Figure 6 | BMI software suite



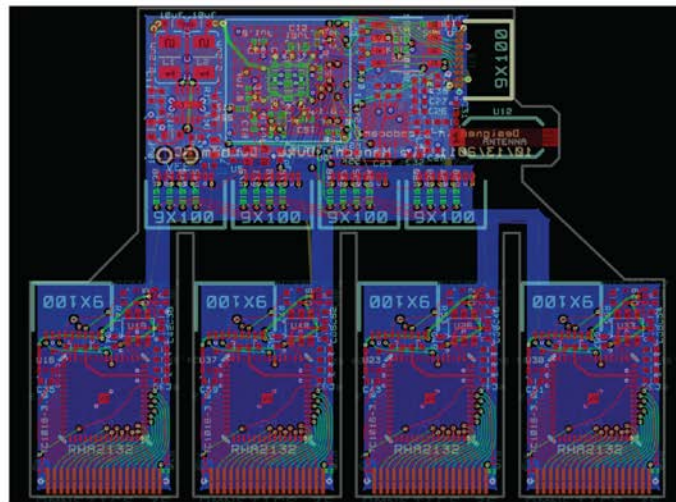
(a) Software workflow for BMI software suite, showing the dynamic integration of signals (LFP, waveform, EMG) recorded from either Plexon or spikes from our wireless system and different behavioral signals to train our adaptive models for BMI experiments. **(b)** Example screenshots of the BMI software suite, showing parameter editing (left) and spiking histogram readouts (right).

Each panel shows the functional block diagram for each module, separated into the digitizer and the transceiver (usually considered in this text as just the transceiver), and the bridge, or receiver.

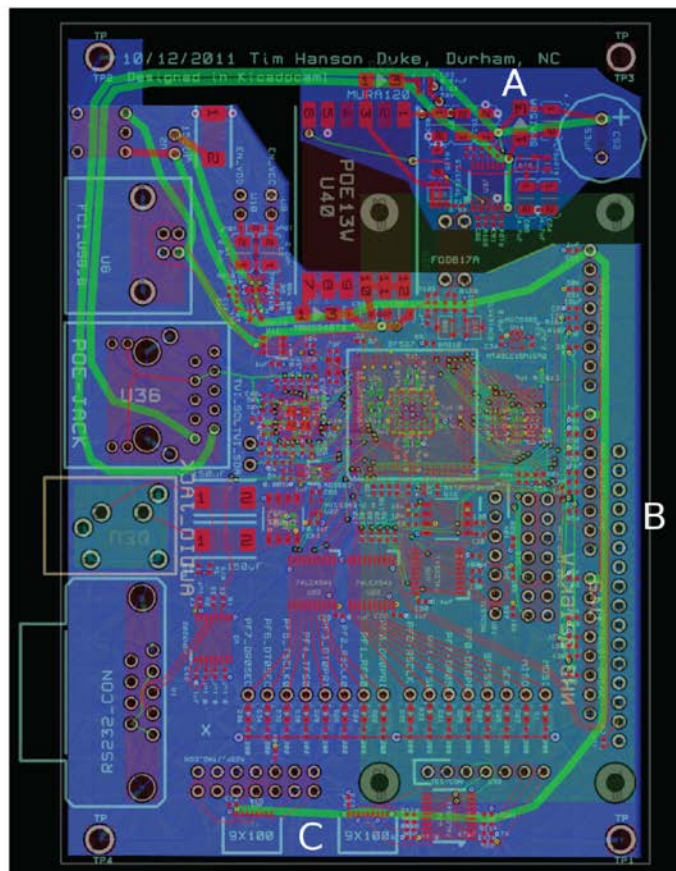


Supplementary Figure 8 | Wireless component trace diagrams

a

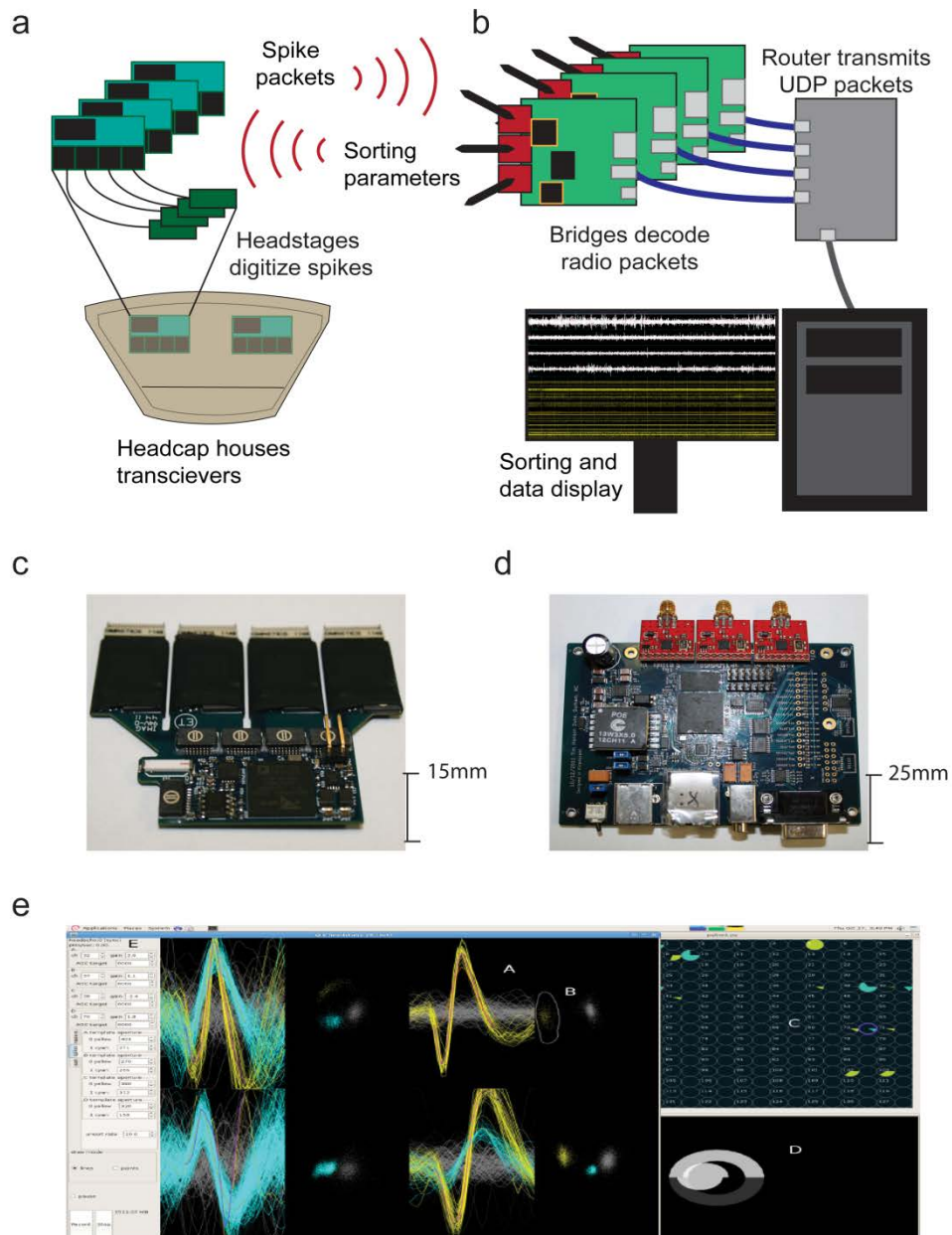


b



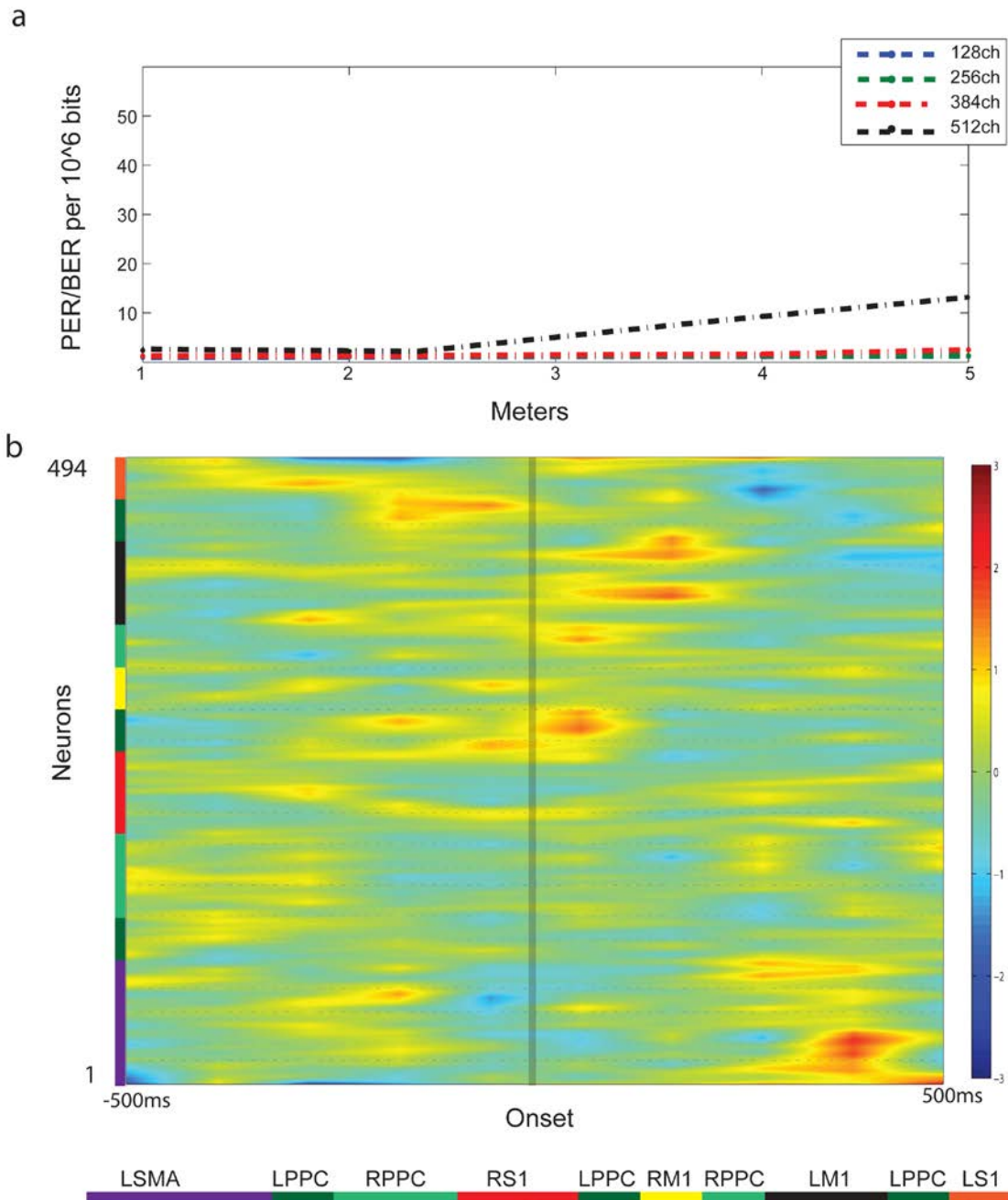
(a) Trace diagrams for the transceiver module connected directly to the headstages (PCBs with slots between them). **(b)** Trace diagram for the bridge module used for interfacing the recordings to the computer client.

Supplementary Figure 9 | Large scale wireless system



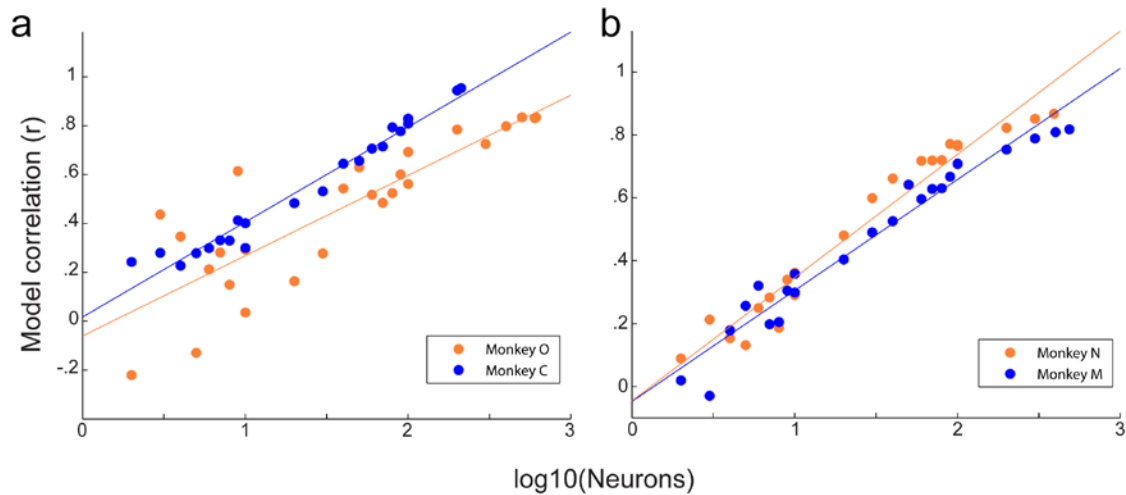
Schematic diagram showing the information flow of the wireless recording system, beginning with spike waveforms to spikes transmitted across ISM radio band to the client. **(a)** Exploded diagram of wireless transceivers located inside the headcap. **(b)** Wireless bridges showing the main components of bidirectional communication and the connections to the client computer. **(c)** Photograph of the radio transceiver with scale. **(d)** Photograph of the wireless bridge, with scale. **(e)** Screenshot of the wireless client with four channels visible, demonstrating the PCA sorting method and graphical user interface.

Supplementary Figure 10 | Wireless scaling



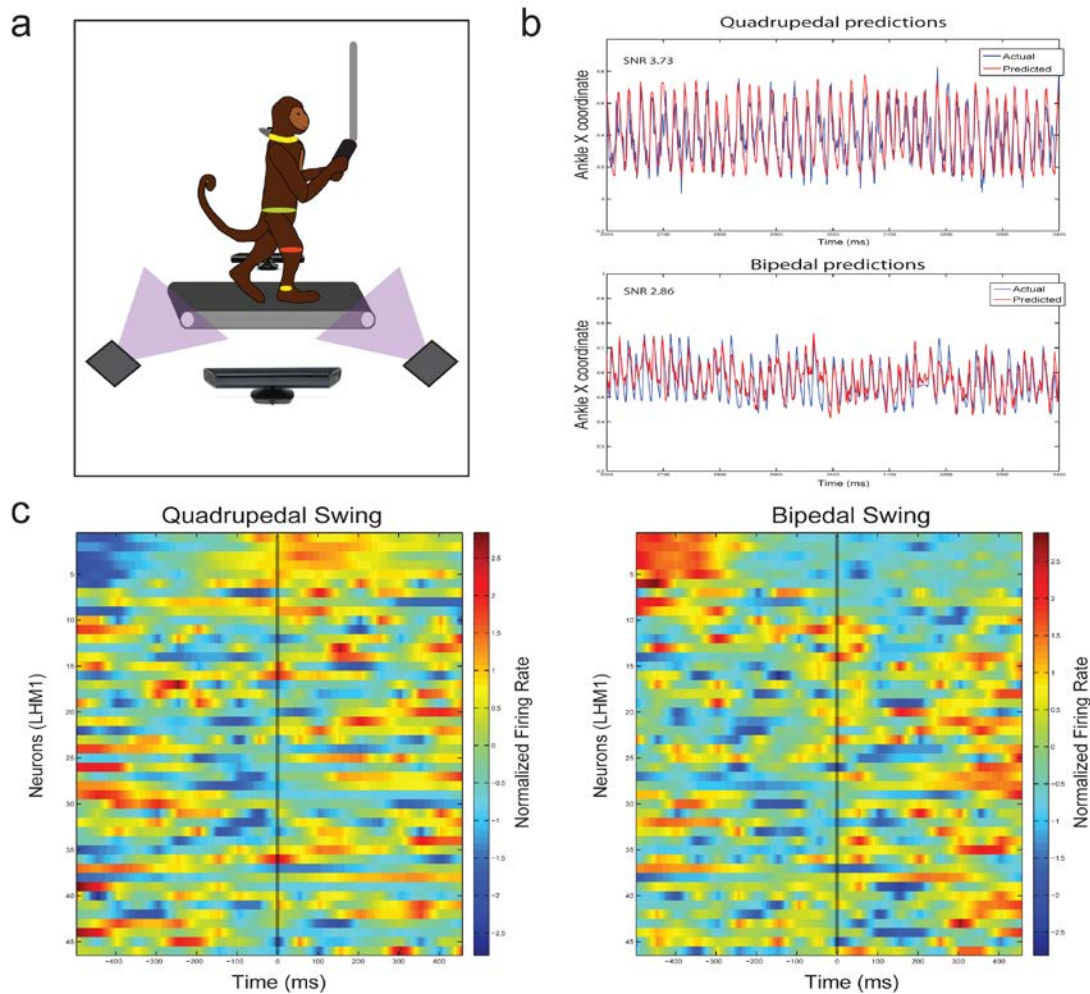
(a) Panel shows packet error rate/bit error rate vs. range per recording module used, as an illustration of scaling performance versus range. **(b)** Spectrogram for the reaching task illustrates 512 channel wireless recordings aligned to onset of reaching. Each unit is identified by color coded area. Units were normalized for the session using z-score. Color legend indicates area of unit location.

Supplementary Figure 11 | Neuron scaling performance metric



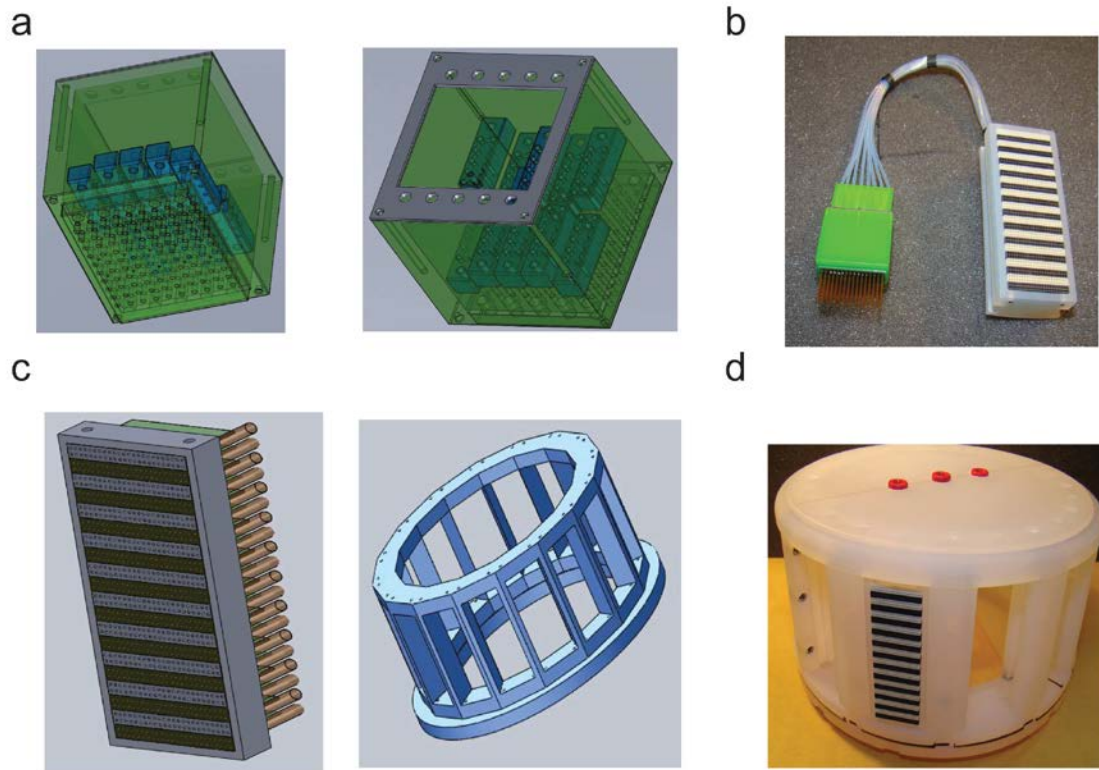
(a) Semi-log dropping curve showing model prediction performance (Unscented Kalman Filter, correlation between prediction model and actual variable) as a base10 logarithmic function of neuron number. This analysis was performed on the x position of the cursor during a center-out task (monkeys O and C). **(b)** Same analysis performed on two monkeys (monkeys M and N) during a bipedal walking trial, with the predicted variable shown as the x position of the ankle.

Supplementary Figure 12 | Wireless recordings during locomotion



(a) Setup of locomotion data acquisition, showing blacklight illumination, cameras, markers for optical tracking, and minimal restraint of the monkey. **(b)** Predictions of the x-coordinate of the right ankle during both bipedal and quadrupedal locomotion. **(c)** PETHs of swing phase for both bipedal and quadrupedal locomotion, calculated from 45 left hemisphere M1 neurons. Neurons are ranked according to maximal activity during onset of the right hindlimb swing phase during quadrupedalism. This indexing is applied to the bipedal swing phase plot.

Supplementary Figure 13 | Next generation arrays



(a) 864 channel array assembly modules. Left panel shows the bottom view with guiding tube density. Right panel shows the top view with a depiction of the movable assemblies. **(b)** Photograph of fully assembled array with guiding tubes. Note the high density connector attached. **(c)** Connector and cap design. Left panel shows schematic for high density connectors. Right panel shows a redesign of the cap which should allow for the management of 10,000 channels without increasing the cap footprint. **(d)** Panel shows photograph of plastic cap mounted with a single connector. The fully assembled cap will host 12 connectors, each with an 864 channel capacity.

Supplementary Tables

Supplementary Table 1. BMI Specifications

Time delay	<1 ms
Time synchronization accuracy	~0.3 ms
Bandwidth	130 ksps
Sampling rate	1-100kHz

Supplementary Table 2. Wireless Specifications

Radio band	ISM (2.54 GHz)
Channels	128 per module
Maximum channels	2560
Sampling rate	31.25 ksps/ch
Sorting	Templates
Templates per channel	2
Power consumption	264 mW
Battery life	>30 hrs
Input referred noise	4.9 μ Vrms
Analog gain	200
ADC resolution	12 bits
LSb of ADC	3.66 μ V
Radio bandwidth	Out: 1.33 Mbps In: 83.3 kbps
Dimensions	Headstages: 1.37 x 2.15 x 0.16 cm Transceiver: 4.3 x 2.1 x 0.34 cm

Supplementary Table 3. Behavioral clustering and classification performance

Algorithm	Performance
K-means	0.928
Expectation Maximization	0.886
SVM (linear kernel)	0.922
SVM (RBF kernel)	0.904

Supplementary Table 4. State-of-the-art wireless recording systems

Reference	No. of Channels	Sampling (ksps)	Data direction	Compression	Radio Band	Band width	Noise (μ Vrms)	ADC (bit)	Power (mW)
Rizk et al., 2007 ¹	96	31.25	Downlink	Raw data	916.5 MHz	1 Mbps	--	12	100
Chestek et al., 2009 ²	96	15.7	Downlink	Raw data	900MHz	345.6 kbps	27.4	10	63.2
Sodagar et al., 2009 ³	64	62.5	Downlink	Spike detection	4MHz	2 Mbps	8	8	14.4
Bonfanti et al., 2010 ⁴	64	20	Downlink	Spike waveform	400MHz	1.25 Mbps	3	8	6.7
Roy & Wang, 2012 ⁵	15	100	Downlink	Raw data	3.05 GHz	200 kbps	5.5	--	30
Borton et al., 2013 ⁶	100	20	Downlink	Raw data	3.8 GHz	24 Mbps	8.6	12	90.6
Miranda et al., 2010 ⁷	32	30	Downlink	Raw data	3.7-4.1 GHz	24 Mbps	3.2	12	142
This work	512*	31.25	Bidirectional	Onboard spike sorting	2.4 – 2.524 GHz (ISM)	2Mbps OTA** / 48Mbps aggregate	4.9	12	264

*With four 128 channel transceivers used.

**Over the air compressed data vs on unit data acquisition.

Supplementary Table 4 References

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5. Roy, S. & Wang, X. Wireless multi-channel single unit recording in freely moving and vocalizing primates. *J. Neurosci. Methods* **203**, 28–40 (2012).
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7. Miranda, H., Gilja, V., Chestek, C. A., Shenoy, K. V. & Meng, T. H. HermesD: A High-Rate Long-Range Wireless Transmission System for Simultaneous Multichannel Neural Recording Applications. *IEEE Trans. Biomed. Circuits Syst.* **4**, 181–191 (2010).